



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2024/2025

Course Instance Codes 3BA1, 4BME1, 1OA2, 4BCT1, 3BMS2, 3BS9, 3FM2
Exams 3rd and 4th year (Arts and Science), Study Abroad

Module Code MA342
Module Topology

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Instructions: **Answer all four questions.**
Each question carries 25 marks.
All workings must be shown.

Duration 2 hours
No. of Pages 3 pages, including this one
Discipline Mathematics

Requirements: No special requirements
Release to Library Yes
Release in Exam Venue Yes
MCQ No
Statistical / Log Tables No
Non-programmable Calculator Yes
Mathematical Tables No
Graph paper No

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- Q1.** (a) Let X be a set.
- (i) Define what is meant by a *topology* on X .
 - (ii) Give an example of a finite set X and two distinct topologies on X . [5 marks]
- (b) Define the *cofinite topology* on a set X . Do the cofinite topology and the discrete topology on the set of integers $\mathbf{Z}$ coincide? Justify your answer. [5 marks]
- (c) (i) Let X be a topological space and let $A \subseteq X$ be a subset. Define the *subspace topology* on A .
- (ii) Let $\mathbf{R}$ denote the set of real numbers endowed with the standard topology. Is the subspace topology on the set of integers $\mathbf{Z}$ discrete? Justify your answer. [5 marks]
- (d) Describe a base for the *lower limit topology* on the set of real numbers $\mathbf{R}$. Show that $[0, 1)$ is both open and closed in the lower limit topology. [5 marks]
- (e) Let X be a topological space, $x \in X$, and $A \subseteq X$.
- (i) Define what it means for a subset N of X to be a *neighbourhood* of x .
 - (ii) Define the *interior* A° of A .
 - (iii) Show that A is a neighbourhood of x if and only if $x \in A^\circ$. [5 marks]
- Q2.** (a) Show that every infinite subset of $\mathbf{R}$ is dense in the cofinite topology on $\mathbf{R}$. [4 marks]
- (b) Define what it means for a topological space to be *compact*. [4 marks]
- (c) Show that every closed subset of a compact topological space is compact. [4 marks]
- (d) Is $\mathbf{R}$ endowed with the lower limit topology compact? Justify your answer. [4 marks]
- (e) (i) Define what it means for a topological space to be *Hausdorff*.
- (ii) Is $\mathbf{Z}$ endowed with the cofinite topology Hausdorff? Justify your answer.
 - (iii) Is $\mathbf{R}$ endowed with the standard topology Hausdorff? Justify your answer. [9 marks]

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- Q3.** (a) Define the *initial topology* on a set with respect to a family of functions. [4 marks]
- (b) Define what it means for a topological space to be *connected*. Give an example each (without justification) of
- (i) a connected space and
 - (ii) a disconnected space. [8 marks]
- (c) (i) Show that $\mathbf{R}$ and $[0, 1]$ are not homeomorphic.
- (ii) Show that $\mathbf{R}^2$ and $\mathbf{R}$ are not homeomorphic.
- (iii) Show that $\mathbf{R}$ and $(0, \infty)$ are homeomorphic. [9 marks]
- (d) Define the *Cantor set*. [4 marks]
- Q4.** (a) (i) Let $f: X \rightarrow Y$ be a continuous surjection between topological spaces. Show that if X is connected, then so is Y .
- (ii) Briefly explain how the preceding part implies the Intermediate Value Theorem from real analysis. [6 marks]
- (b) Let X and Y be topological spaces. Write down a base for the *product topology* on $X \times Y$. [4 marks]
- (c) Let X be a topological space and let $\sim$ be an equivalence relation on X . Define the *quotient topology* on $X/\sim$. [4 marks]
- (d) Briefly explain why $\mathbf{S}^1$ is a quotient of $[0, 1]$. [5 marks]
- (e) Let X and Y be topological spaces.
- (i) Let $f: X \rightarrow Y$ and $g: X \rightarrow Y$ be continuous functions. Define what it means for f and g to be *homotopic*.
 - (ii) Let $f: \mathbf{R}^m \rightarrow \mathbf{R}^n$ and $g: \mathbf{R}^m \rightarrow \mathbf{R}^n$ be continuous functions. Show that f and g are homotopic. [6 marks]